Score Conditioning and Static Threshold Calibration in Multivariate Time-Series Anomaly Detection: An Empirical Study

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***Abstract*—Automated anomaly thresholds can generate excessive alarms when telemetry scores change between calibration and deployment. We compare latent predictive, observation-space, and simple bounded scores under chronological held-out calibration on 33 streams from seven datasets; a 19-stream non-GHL cohort supports the five-seed controls. On that cohort, the latent predictor has lower nominal false-positive rates than two observation-space reconstruction baselines. Still, a bounded, dimensionality-matched principal-component control and a bounded observation-space TCN control also produce light residual tails. The TCN control, nevertheless, has higher calibration false-positive rates, indicating that tail shape and threshold transfer are distinct. Causal event-recall curves suggest a different ordering when plotted against realized false-positive rate, but paired interpolation on a common observed range covers few streams and does not establish an aggregate advantage; buffered attribution is analyzed separately. Point-F1 and precision-recall differences remain imprecise, not equivalent. Static peaks-over-threshold calibration substantially overshoots its target risk under this short-block protocol. Inter-block score drift and scarce tail exceedances are both plausible contributors; their relative causal roles remain unresolved. Three physics-inspired variants show no reliable aggregate detection gain. These findings emphasize the use of matched-score controls, the reporting of realized alarm rates, and the limits of what nominal-budget curves can establish.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Threshold calibration, Extreme value theory, Representation learning.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable digital healthcare [1], [2]. In remote sensing deployments — NASA SMAP, MSL rovers, industrial IoT facilities, wearable biometric arrays — dozens to hundreds of channels concurrently track power buses, thermal gradients, vibrations, and kinematics [3], and faults must be identified before hardware damage occurs.

A. Challenges in Modeling Multi-Sensor Telemetry

Multi-sensor telemetry reflects a physical state trajectory $\dot{x}(t) = \mathbf{f}(x(t))$, yet modeling dynamics in raw observation space faces four obstacles: estimating derivatives $\dot{x} \approx \Delta x/\Delta t$ on noisy discrete streams amplifies high-frequency noise; telemetry captures only a noisy projection of unmeasured states, violating full-state observability; in large arrays ($K \geq 25$) cross-channel nonlinear interactions resist rigid functional forms; and discrete operational mode transitions (orbital day/night cycling, thruster firings, gait freezing) induce continuous distribution drift.

B. Limitations of Deep Anomaly Detectors

Reconstructive models score observation-space errors, whereas latent predictors and low-dimensional controls use differently scaled, often bounded scores [3]–[5]. Comparing their alarm rates at a common nominal threshold risk does not separate scoring geometry from discriminative quality. Nor does a static threshold fitted on one nominal block guarantee the same false-positive rate on a later block. Whether high-frequency noise specifically consumes reconstruction capacity, or synthetic perturbations in contrastive methods such as NCAD [6] distort decisions, remains untested here.

C. Investigating Latent Prediction and Inductive Priors

We compare score conditioning, detection, and threshold transfer separately. A JEPA predicts target representations from context [7], [8]; observation-space models, a matched-backbone control, a LayerNorm-bounded observation-space TCN, and a 32-dimensional PCA forecast with normalized gated scoring test alternative explanations. We also evaluate spectral entropy, latent second moment, and potential-flow variants as exploratory physics-inspired controls. These experiments do not by themselves isolate a noise-removal mechanism.

D. Contributions

This work makes four principal contributions:

1) Disentangling Alarm Rates and Score Quality: On 19 non-GHL streams, the nominal held-out false-positive rate is 7.6% for TS-JEPA versus 30% for TimesNet, but bounded PCA and bounded observation-space TCN controls also produce light residual tails, and the TCN control does not preserve

low calibration FPR. The larger JEPA event-recall area over *nominal design budgets* under causal attribution does not establish dominance at matched realized FPR: few streams span the same realized-FPR range, and paired comparisons are inconclusive. Buffered and causal results are reported separately.

2) Null Results for Physics-Inspired Variants: Across the 33-stream SPOT cohort, the tested regularizers provide no reliable aggregate detection gain. A Potential-Flow Point-F1 difference has an unadjusted p=0.033 but does not justify a multiplicity-adjusted superiority or inferiority claim.

3) Conditional Calibration Failure: Static POT thresholds substantially exceed their nominal design risk under short held-out calibration. Score-distribution differences across blocks correlate with false alarms, but small effective exceedance counts and observational confounding prevent causal attribution to drift alone.

4) Reproducible Multi-Seed Benchmark: The SPOT benchmark spans 33 streams from seven datasets (19 non-GHL streams with five seeds; 14 GHL streams with fewer seeds). The score-dump operating-curve cohort is narrower: 19 buffered and 18 causal non-GHL streams. We report paired seed coverage and dataset-level uncertainty for the matched-realized-FPR analysis.

Supplementary material contains proofs, extended diagnostics, and case-study details.

II. Related Work

A. Dynamical Systems, Reconstruction, and Forecasting Detectors

Physical modeling of multi-sensor telemetry frames system behavior as continuous trajectories $\dot{x}(t) = \mathbf{f}(x(t))$ [9]. Neural ODEs [10] parameterize velocity fields in raw observation space, where integrating through high-frequency sensor noise induces numerical stiffness and jitter sensitivity. Koopman theory and DMD [11]–[13] instead learn linear operators on observable embeddings; physical disruptions then manifest as spectral decoherence. Our Operator-Entropy variant transfers this idea to a latent transition operator regularized by spectral entropy during training.

The predominant detection approach learns nominal patterns via raw-signal reconstruction: LSTM prediction with dynamic thresholding for NASA telemetry [3], association-discrepancy attention in Anomaly Transformer [4], 2D FFT-periodicity decomposition in TimesNet [5], and adversarial sub-series attention in TranAD [14]. These share a vulnerability: point-level losses bias networks toward fitting stochastic noise, and sustained anomalies contaminate the context window. These errors accumulate residual variance across irrelevant channels in high-dimensional arrays.

B. Threshold Calibration, Contrastive, and Self-Supervised Detection

Unsupervised detectors require an automated rule that converts residual scores into binary alarms without test labels. Non-parametric dynamic thresholding [3] and Streaming Peaks-Over-Threshold (SPOT) [15]—fitting a Generalized Pareto tail above a high quantile under design risk q—are standard; however, EVT's asymptotic guarantees assume i.i.d. tail samples, which telemetry autocorrelation and operational drift violate.

To avoid reconstructing noise, contrastive methods learn discriminative representations: NCAD [6] trains against synthetic anomalies, yielding lower calibrated detection than predictive models (0.058 vs. 0.160 macro Point-F1, Supplementary Table S28). DCdetector [16] maximizes cross-scale dual-attention consistency. Joint-Embedding Predictive Architectures [7], [8], [17] instead predict target representations in latent space, with non-contrastive collapse prevention [18]–[20]. Time-series extensions [21] evaluate ranking or post-hoc threshold sweeps without studying how latent residuals behave under automated tail calibration—the gap this work addresses.

C. Physics-Inspired Inductive Biases

PINNs [22] and Hamiltonian/Lagrangian networks [23], [24] embed dynamics as hard constraints; their stiff competing gradients [25] and closed-form-PDE requirements fit poorly to uncalibrated multi-sensor telemetry. We instead test whether *inspired* structures—spectral entropy, second-moment closure, conservative potential flow—improve latent residuals.

D. Terminology and Positioning

We distinguish *physics-inspired* structures (exploratory regularizers; latent coordinates are not claimed to be physical state variables) from *physics-informed* constraints enforcing known governing equations; *causal preprocessing* (features computed strictly from history) from *causal decision-time access* (alarm latency); and transition-operator non-degeneracy from representation anti-collapse. This investigation bridges joint-embedding predictive learning and multivariate anomaly detection: three physics-inspired regularizers are evaluated against unconstrained latent controls, and EVT calibration is evaluated on strictly held-out nominal telemetry to quantify the gap between design risk and realized false-alarm rates.

III. Multi-Sensor IoT and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams sourced from NASA spacecraft archives [3], the UCI Machine Learning Repository [26], [27], TimeEval [28], and mTSBench [29]: 33 streams are evaluated under the primary multi-seed held-out calibration protocol and 12 reserved for operational case studies. Streams span four physical domains: *spacecraft telemetry* (SMAP Earth-observation `A-1`/`A-2`/`A-7`/`D-3`/`P-3`/`E-1` power, thermal, and attitude subsystems; MSL rover `C-1`/`C-2`/`D-14`/`D-15`/`D-16`/`M-1` actuator and environmental channels, with two short-nominal streams reserved; SWAN-SF solar magnetograms [30]); *wearable kinematics* (six Daphnet FoG streams [26]: 64 Hz tri-axial accelerometry for Freezing-of-Gait events, so S=64 is a 1-s horizon); *industrial processes* (14 GHL gasoil heating-loop streams, the Genesis pick-and-place testbed, and GECCO water-distribution

chemistry/hydrology at 1-minute resolution); and *smart infrastructure* (Room-Occupancy environmental sensing [31], six CICIDS edge network-flow streams, CalIt2 people-flow [27], and Metro I-94 highway traffic).

Each stream provides nominal training sequences and labeled anomaly test sequences; channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatline phases. We evaluate the standardized subsets studied in the literature [3], [28] (of 55 SMAP and 27 MSL archive entities) for strict matching and reproducible runtime.

To eliminate circular in-sample calibration, each stream's nominal sequence is partitioned chronologically into training (65%), calibration (17.5%), and held-out evaluation (17.5%) blocks separated by an isolation gap (Section V-A). The held-out protocol spans 33 streams across seven datasets (6 Daphnet, 14 GHL, 4 MSL, 6 SMAP, Genesis, Metro, SWAN); the remaining 12 streams are reserved as operational case studies — CalIt2, two Room-Occupancy, two short-nominal MSL, GECCO, and six CICIDS — because their nominal blocks are too short for disjoint partitions or because multi-seed stride-1 evaluation over 10^6 steps was prohibitive. The dyadic front-end ablation (Table Ib) uses a separate five-stream panel that additionally draws one SMD channel outside this manifest.

Two caveats bound interpretation. SMAP and MSL contain documented mislabeled segments and near-trivial anomalies [32]; their per-dataset numbers are upper bounds on detectability. Entity coverage is also incomplete — 6 of 55 SMAP and 4 of 27 MSL channels, selected by nominal-block length and stride-1 runtime constraints, not labels — so claims apply to the evaluated entities, not the full archives.

Supplementary Table S15 specifies the suite: 2,079,507 nominal training and 2,986,574 test timesteps (>5.06M observations), with per-dataset domain, channel dimension K, sequence lengths, contamination ratio $\pi\%$, anomaly typologies, and partition protocols.

A machine-readable manifest (`reports/stream_manifest.csv`, generated by `scripts/build_stream_manifest.py`) enumerates every stream's identifier, channel dimension, split lengths, held-out anomaly counts, benchmark role, and evaluated seed set as recorded in the released artifacts. All counts in this section, the abstract, and the results tables reconcile against it: 33 primary held-out streams (19 non-GHL at five seeds, 14 GHL at three to five seeds depending on size) and 12 case-study streams.

IV. Methodology: Latent Representation Forecasting and Structured Inductive Priors

This section formalizes the mathematical formulation, latent predictors, and inductive biases of the proposed framework.

A. Problem Formulation and Latent Space Prediction

Let $\mathbf{X} \in \mathbb{R}^{T\times K}$ denote a multivariate stream partitioned into context $x_{\text{ctx}} = \mathbf{X}[t-C:t,:] \in \mathbb{R}^{C\times K}$ and suspect horizon $x_{\text{tgt}} = \mathbf{X}[t:t+S,:] \in \mathbb{R}^{S\times K}$. Reconstructive baselines (TimesNet, TranAD, Anomaly Transformer) minimize pointwise error $\min_\theta \mathbb{E}\|\mathbf{X} - \mathcal{D}_\theta(\mathcal{E}_\theta(\mathbf{X}))\|_F^2$, which can weight channels with different noise scales unequally; whether this consumes model capacity is not tested here. We instead adopt the JEPA framework [7]: a context encoder E_θ maps $x_{\text{ctx}} \to z_{\text{ctx}} \in \mathbb{R}^D$ and an EMA momentum target encoder E_ϕ maps $x_{\text{tgt}} \to z_{\text{tgt}}$, with $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$, $m_k \in [0.996, 0.9995]$. The objective forecasts target representations in latent space and tests whether structured priors condition prediction residuals for tail calibration.

B. Latent Predictor Architectures and Dynamical Inductive Priors

To examine whether structured dynamical inductive biases stabilize representation forecasting, we evaluate an unconstrained latent predictor alongside three candidate formulations inspired by physical dynamical principles:

1) Operator-Entropy JEPA (Spectral Dynamics & Modal Dispersion): Motivated by Koopman operator theory [33], [34], where nonlinear trajectories are mapped via linear operators on observable functions, we parameterize the predictive latent transition via a learned operator $\mathbf{K} \in \mathbb{R}^{D\times D}$: $\hat{z}_{\text{tgt}} = \mathbf{K}z_{\text{ctx}}$. To prevent collapse onto a low-rank subspace (dimensional collapse) or to zero (energy collapse), the regularizer decouples spectral uniformity from total operator energy.

Let $\sigma_1 \geq \cdots \geq \sigma_D \geq 0$ denote singular values of $\mathbf{K}$, $\lambda_i = \sigma_i^2$ eigenvalues of $\mathbf{K}\mathbf{K}^\top$, and $\mathcal{E}(\mathbf{K}) = \|\mathbf{K}\|_F^2 = \sum_i \lambda_i$ total energy. For $\mathcal{E}(\mathbf{K}) > 0$, the scale-free modal distribution is $p_i = \sigma_i^2/\|\mathbf{K}\|_F^2$. The Von Neumann spectral entropy is $\mathcal{H}_{\text{spectral}}(\mathbf{K}) = -\sum_{i:p_i>0} p_i \log p_i$ (with $0\log 0 = 0$). The *Spectral Entropy Deficiency* measures Kullback-Leibler divergence from the uniform equipartition $\mathbf{u} = [1/D, \ldots, 1/D]^\top$:

$$\Delta\mathcal{H}(\mathbf{K}) = \log D - \mathcal{H}_{\text{spectral}}(\mathbf{K}) = \mathcal{D}_{\text{KL}}(\mathbf{p} \,\|\, \mathbf{u}) \geq 0, \quad (1)$$

where $\Delta\mathcal{H} = 0$ if and only if all modes carry equal energy ($\sigma_1 = \cdots = \sigma_D$), while dimensional collapse to rank 1 yields $\Delta\mathcal{H} = \log D$.

To penalize energy collapse ($\|\mathbf{K}\|_F \to 0$) without explosive gain, we impose a non-negative Bregman energy barrier relative to nominal isometric scale $\mathcal{E}_0 = \gamma D$ ($\gamma = 1.0$):

$$\Phi_{\text{energy}}(\mathbf{K}) = r_{\text{safe}}(\mathbf{K}) - 1 - \log r_{\text{safe}}(\mathbf{K}) \geq 0, \quad (2)$$

where $r_{\text{safe}}(\mathbf{K}) = \max(\|\mathbf{K}\|_F^2/(\gamma D), \epsilon)$ with numerical floor $\epsilon = 10^{-6}$. The floored barrier caps penalty at $\approx 12.8\,\beta_{\text{norm}}$ rather than diverging. With $\mathbf{K} \sim \mathbf{I} + \mathcal{N}(\mathbf{0}, 0.01^2\mathbf{I})$, $r_{\text{safe}} \approx 1$ and training begins with vanishing barrier penalty. The composite regularizer and objective are:

$$\mathcal{R}_{\text{spectral}}(\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K}) + \beta_{\text{norm}}\Phi_{\text{energy}}(\mathbf{K}), \quad (3)$$

$$\mathcal{L}_{\text{Op-Entropy}} = \|\mathbf{K}z_{\text{ctx}} - z_{\text{tgt}}\|_2^2 + \lambda_{\text{ent}}\mathcal{R}_{\text{spectral}}(\mathbf{K}), \quad (4)$$

with $\beta_{\text{norm}} = 1.0$ and $\lambda_{\text{ent}} = 0.5$. All shared representation components are held fixed across models and controls. $\mathcal{R}_{\text{spectral}}(\mathbf{K})$ acts on the operator matrix; it does not analytically prevent encoder representation collapse, which is mitigated empirically by EMA target dynamics and stop-gradient operations [8], [35].

Proposition 1 (Operator Non-Triviality and Barrier Behavior)**.** *Let* $\mathcal{L}(\mathbf{K}) = \frac{1}{B}\sum_{b=1}^{B}\|\mathbf{K}z_{ctx}^{(b)} - z_{tgt}^{(b)}\|_2^2 + \lambda_{ent}\mathcal{R}_{spectral}(\mathbf{K})$ *with* $\lambda_{ent} > 0, \beta_{norm} > 0$*, non-trivial target batch*

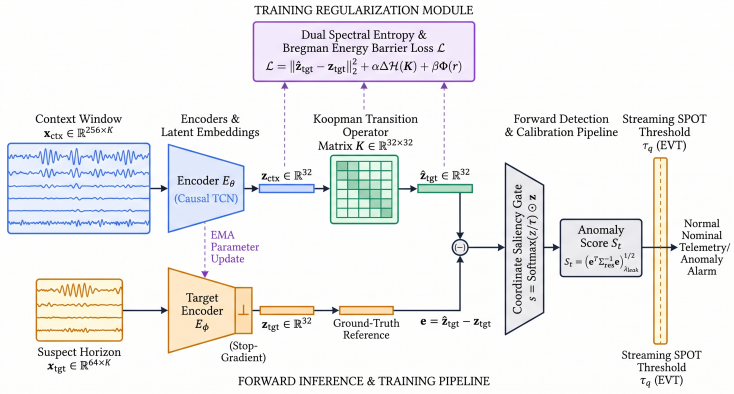


Fig. 1. Operator-Entropy JEPA architecture: multiscale context ($\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{256\times K}$) and suspect ($\mathbf{x}_{\text{tgt}} \in \mathbb{R}^{64\times K}$) windows are mapped into latent space via online (E_θ) and EMA momentum (E_ϕ) encoders with a stop-gradient barrier ($\perp$). A linear Koopman transition operator ($\mathbf{K} \in \mathbb{R}^{32\times32}$) forecasts the target representation $\hat{\mathbf{z}}_{\text{tgt}} = \mathbf{K}\mathbf{z}_{\text{ctx}}$, regularized by dual Von Neumann spectral entropy and a logarithmic Bregman energy barrier against collapse. Residual discrepancies pass through a coordinate saliency gate to yield anomaly score S_t calibrated via streaming SPOT.

$\frac{1}{B}\sum_{b=1}^{B} \|z_{tgt}^{(b)}\|_2^2 = E_{tgt} > 0$, *and non-degenerate context–target coupling* $\frac{1}{B}\sum_{b=1}^{B} z_{tgt}^{(b)} z_{ctx}^{(b)\top} \neq \mathbf{0}$. *Then: (i) under the floored barrier* ($\epsilon = 10^{-6}$)*, the energy penalty is bounded* ($\leq 12.8\,\beta_{norm}$) *yet* $\mathbf{K} = \mathbf{0}$ *is strictly not a local or global minimizer; and (ii) under unfloored idealization* ($\epsilon \to 0^+$)*,* $\lim_{\|\mathbf{K}\|_F \to 0} \mathcal{L}(\mathbf{K}) = +\infty$.

(Proof in Supplementary Section S-1.)

Remark (Operator Non-Degeneracy vs. Representation Collapse): Proposition 1 applies to the transition operator $\mathbf{K}$ and does not analytically prevent encoder collapse: if E_θ maps all inputs to a constant, $\mathbf{K} = \mathbf{I}$ attains zero loss. Preventing representation collapse rests empirically on stop-gradient momentum dynamics.

2) Auxiliary Quadratic Second-Moment Head: Inspired by Reynolds decomposition in fluid turbulence, which separates mean flow from velocity fluctuations, we evaluate an auxiliary quadratic second-moment head in latent space. The observed matrix $\boldsymbol{\Sigma}_{\text{obs}} = (\mathbf{P}z_{\text{tgt}})(\mathbf{P}z_{\text{tgt}})^\top + 10^{-3}\mathbf{I} \in \mathbb{S}_{++}^{d_s\times d_s}$ is an instantaneous rank-1 projected outer product plus ridge penalty, not a physical turbulent fluctuation covariance.

A two-layer MLP predicts mean target latent $\hat{z}_{\text{tgt}}$ from z_{ctx}, and linear projection $\mathbf{P} \in \mathbb{R}^{d_s\times D}$ ($d_s = 16$) yields $\boldsymbol{\Sigma}_{\text{obs}}$. The closure head $h_\psi(z_{\text{ctx}})$ outputs lower-triangular $\mathbf{L} \in \mathbb{R}^{d_s\times d_s}$ with positive diagonal $L_{ii} = \text{softplus}(a_{ii}) + \epsilon$, reconstructing predicted covariance:

$$\boldsymbol{\Sigma}_{\text{pred}} = \mathbf{L}\mathbf{L}^\top + 10^{-3}\mathbf{I} \in \mathbb{S}_{++}^{d_s\times d_s}. \tag{5}$$

The training objective is:

$$\mathcal{L}_{\text{Reynolds}} = \text{MSE}(\hat{z}_{\text{tgt}}, z_{\text{tgt}}) + 0.1\ \text{MSE}(\boldsymbol{\Sigma}_{\text{pred}}, \boldsymbol{\Sigma}_{\text{obs}}). \tag{6}$$

At test time, the anomaly score adds the second-moment error: $\mathcal{S}(t) = g(z_{\text{tgt}} - \hat{z}_{\text{tgt}}) + 0.5\,g(\text{vec}(\boldsymbol{\Sigma}_{\text{obs}} - \boldsymbol{\Sigma}_{\text{pred}}))$. Downstream performance is evaluated in Section VII-B and Table I(a).

3) Potential-Flow JEPA (Conservative Field Theory and Flow Matching): Under Helmholtz-Hodge decomposition, a smooth field splits into a curl-free potential gradient and a solenoidal component: $v = -\nabla\Phi + \nabla\times A$. Dissipative gradient flows provide theoretical motivation:

Proposition 2 (Idealized Continuous Gradient Flow Motivation (Non-Binding))**.** *Let* $\dot{z} = -\nabla\Phi(z)$ *be an autonomous gradient flow on* $\mathbb{R}^D$ *with* $\Phi \in C^2(\mathbb{R}^D, \mathbb{R})$*. There exist no non-constant periodic orbits.*

(Proof in Supplementary Section S-1.)

a) Velocity Field, Multi-Subspace Routing, and Flow Matching: The continuous field is $\hat{v}(z_t, t, z_{\text{ctx}}, p_{\text{regime}}) = -\nabla_{z_t}\Phi_\psi$, where Φ_ψ is a three-layer residual MLP conditioned on state z_t, diffusion time $t \in [0,1]$, context z_{ctx}, and regime projection p_{regime}, with tangent projection $v_\perp = v - \langle u, v\rangle u$ ($u = z_t/\|z_t\|_2$). Operational regimes use $K_r = 4$ orthonormal subspace projections $\mathbf{U}_k \in \mathbb{R}^{D\times d}$ (d=8, D=32), selected by maximal captured context energy $p_{\text{regime}} = \mathbf{U}_{k^*}\mathbf{U}_{k^*}^\top z_{\text{ctx}}$ with separation loss $\mathcal{L}_{\text{subspace}} = \frac{2}{K_r(K_r-1)}\sum_{i<j}\|\mathbf{U}_i^\top\mathbf{U}_j\|_F^2$.

Training uses conditional flow matching (OT-CFM) along linear paths between noise $z_{\text{noise}} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and z_{tgt}: $z_t = (1-t)z_{\text{noise}} + t z_{\text{tgt}}$, with loss

$$\mathcal{L}_{\text{Pot-Flow}} = \mathbb{E}_{t, z_{\text{noise}}}\left[\|\hat{v}(z_t, t, z_{\text{ctx}}, p_{\text{regime}}) - (z_{\text{tgt}} - z_{\text{noise}})\|_2^2\right]. \tag{7}$$

At inference, the detector queries the field at the trajectory midpoint:

$$\mathbf{e}(t) = \hat{v}(0.5 z_{\text{tgt}},\ 0.5,\ z_{\text{ctx}},\ p_{\text{regime}}) - z_{\text{tgt}}. \tag{8}$$

Discrete midpoint queries on conditional velocity fields do not integrate an autonomous continuous trajectory and do not inherit autonomous non-periodic guarantees; Proposition 2 provides idealized continuous motivation and does not bind discrete scoring. Parameterizing $\hat{v}$ as a spatial gradient imposes a curl-free bias ($\nabla\times\hat{v} \equiv \mathbf{0}$) with vanishing circulation $\oint_{\mathcal{C}} \hat{v}\cdot d\mathbf{r} = 0$, dynamically penalizing cyclic orbits as anomalous and degrading detection on periodic kinematics (Section VII-B).

C. Anomaly Scoring and Operational Detection Latency

During inference, context windows $x_{\text{ctx}} = \mathbf{X}[t-C : t, :]$ and suspect windows $x_{\text{tgt}} = \mathbf{X}[t : t+S, :]$ are processed at stride 1. For TS-JEPA, Operator-Entropy, and Potential-Flow, the score gates the latent discrepancy through $g(\cdot)$ (Section IV-D1): $\mathcal{S}(t) = g(\hat{z}_{\text{tgt}}(t) - z_{\text{tgt}}(t))$. For the second-moment head (Reynolds-Stress), $\mathcal{S}(t) = g(z_{\text{tgt}} - \hat{z}_{\text{tgt}}) + 0.5\,g(\text{vec}(\boldsymbol{\Sigma}_{\text{obs}} - \boldsymbol{\Sigma}_{\text{pred}}))$. All shared representation stages are held fixed across variants and ablation controls (Section VI-A).

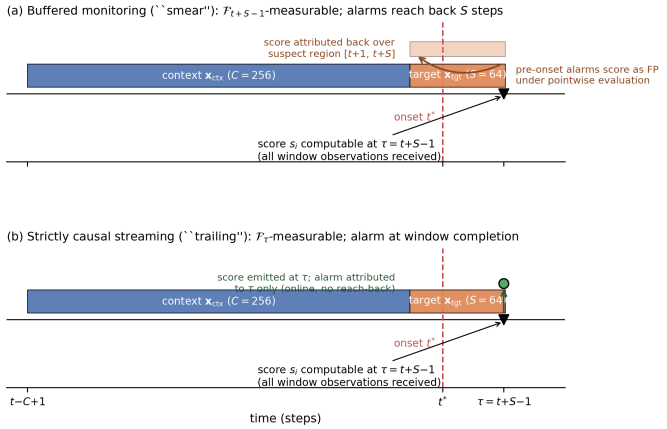


Fig. 2. Score-emission and alarm-attribution timeline. Every score s_i is computable only at window completion $\tau = t+S-1$. (a) Buffered attribution assigns s_i retrospectively to $[t+1, t+S]$; (b) causal attribution emits s_i at τ only.

Three-Part Operational Causality Framework: To resolve the methodological confounding between data availability, decision latency, and evaluation attribution, we formalize detection under three operational criteria (Fig. 2):

(a) **Data causality ($\mathcal{F}_\tau$-measurability):** At decision time $\tau = t + S - 1$, the model observes only history $\mathbf{X}_{:\tau} = \mathbf{X}[t - C : \tau, :]$. Neither encoder accesses future observations beyond τ.

(b) **Score-emission latency ($S - 1 = 63$ steps):** Because suspect representation z_{tgt} encodes the horizon $\mathbf{X}[t : t + S, :]$, window score s_t cannot be computed until timestep $\tau = t + S - 1$. This imposes an inherent emission latency of $S - 1 = 63$ steps: 0.98 s on Daphnet (64 Hz), 10.5 min on SMAP, MSL, and GHL (0.1 Hz), 63 min on GECCO (1 sample/min), and 63 h on Metro (1 sample/h).

(c) **Alarm attribution convention:** We distinguish two attribution regimes: (1) *strictly causal trailing attribution*, where score s_t is emitted and evaluated strictly at τ, providing zero-lookahead online monitoring; and (2) *retrospective buffered smear attribution*, where s_t is attributed across the entire suspect window $[t+1, t+S]$. Under unadjusted point evaluation, buffered alarms in $[t^* - S, t^*]$ register as false alarms: buffering provides fixed-lag detection, not early warning. The two regimes represent distinct operational realities and are reported separately throughout (Table II, Supplementary Table S31).

D. Shared Internal Inductive Biases

To stabilize learning across high-dimensional, noisy telemetry, all three architectures incorporate three inductive mechanisms:

1) Coordinate Saliency Gating and the Auxiliary Channel-Gate Variant: On high-dimensional streams ($K \geq 25$), faults frequently originate in a localized subsystem while the remaining channels exhibit ambient noise, diluting localized anomalies in isotropic representations. The deployed detector mitigates this in latent space via an anti-drowning coordinate saliency gate on error vectors $\mathbf{e} = z_{\text{tgt}} - z_{\text{pred}} \in \mathbb{R}^D$:

$$\mathcal{D}_{\text{sal}}(\mathbf{e}) = (1 - \alpha)\|\mathbf{e}\|_2 + \alpha\sqrt{D\sum_{d=1}^{D} w_d e_d^2}, \tag{9}$$

$$w_d = \frac{\exp(|e_d|/\tau)}{\sum_{j=1}^{D}\exp(|e_j|/\tau)},$$

amplifying sparse coordinate deviations by up to $\sqrt{D}$ while remaining exactly $\|\mathbf{e}\|_2$ under isotropic errors. This is the discrepancy gate used in every experiment: input channels are encoded unweighted, and the gate acts only on the D-dimensional predictive residual.

a) Auxiliary Input-Channel Gate (Evaluated Variant): We also evaluate a strictly causal *Causal Channel Saliency Gate* as an auxiliary variant (full specification, boundary-condition verification, and the fault-type study in Supplementary Section S-3): each channel is weighted by the sigmoid of its standardized context-window variance share, $g_k = \sigma((s_k - \bar{s})/(\sigma_s + \epsilon))$, applied symmetrically to context and target encoders so that $\partial g_k/\partial \mathbf{x}_{\text{tgt}} \equiv \mathbf{0}$ (zero target leakage, with an explicit identity short-circuit at $K{=}1$). Two boundary conditions apply: quiet faults (flatlines, stuck values) cannot be isolated by variance amplification — the fault-type study confirms they are weakly detected under *every* gating regime (recall 0.02–0.07) — and abrupt regime transitions transiently perturb variance shares until the context window refills.

b) Zero-Leakage Data Provenance and Temporal Isolation: We enforce strict temporal isolation across the training, nominal-calibration, and held-out evaluation partitions (Section V-A): standardization statistics come only from the training partition, EVT tail parameters (u, σ, ξ) only from the calibration partition, and evaluation proceeds chronologically with frozen weights and thresholds. The auxiliary gate's weights depend only on x_{ctx}; the deployed pipeline weights inputs uniformly.

2) Multiscale Frequency Front-End: Anomalies span temporal scales from sensor spikes to thermal drifts; the encoder front-end therefore splits each channel into $J{=}4$ smooth dyadic shells on an $N_{\text{fft}} = 256$ rFFT grid, normalized bin-wise into an exact amplitude partition of unity, $\sum_j \Psi_j[\omega_m] \equiv 1$ (measured deviation $< 1.2 \times 10^{-7}$; construction details in Supplementary Section S-4). Each shell feeds a causal 1D convolution ($k \in \{15, 9, 5, 3\}$ coarse-to-fine), recombined by learned scale routing. Because the shells overlap smoothly at band edges, the front-end is *less* frequency-localized than standard wavelet banks: per-band in-octave spectral mass is $\{0.50, 0.33, 0.20, \approx 1.00\}$ versus $\{0.78, 0.50, 0.55, 0.82\}$ (Haar) and $\{0.86, 0.63, 0.64, 0.87\}$ (DB4) (`reports/lp_filterbank_diagnostics.csv`). We therefore describe it as a Fourier-domain multiscale front-end, not an exact Littlewood–Paley octave decomposition, and measure its empirical contribution by matched ablation (Table Ib).

3) Kurtosis-Adaptive Extreme Value Theory (EVT) and Calibration Realities: Binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_t)$ are generated via Streaming Peaks-Over-Threshold (SPOT) [15]:

excesses over a high threshold u (98th percentile) asymptotically follow a Generalized Pareto Distribution (GPD) with scale σ and shape ξ:

$$\tau = u + \frac{\sigma}{\xi}\left[\left(\frac{qN}{N_u}\right)^{-\xi} - 1\right], \tag{10}$$

where $q = 10^{-3}$ denotes the design risk probability, N is total calibration samples, and N_u is the count of excesses. To account for varying tail heaviness across diverse sensor regimes, we modulate the threshold via an empirical heuristic factor governed by the sample excess kurtosis γ_2 of calibration residuals:

$$\tau_{\text{eff}} = \tau \cdot \left[1 + \frac{1}{6}\tanh(\gamma_2)\right]. \tag{11}$$

On heavy-tailed distributions ($\gamma_2 > 0$), τ_{eff} rises to suppress volatile-noise false alarms; on light-tailed ones, it tightens for sensitivity.

We emphasize that this is an engineering heuristic for pre-asymptotic finite-sample tails, not an asymptotic EVT theorem. In the controlled calibrator ablation (Section VII-D, Table III) it does not improve out-of-sample risk control: on held-out data the multiplier raises TS-JEPA nominal FPR from 5.26% to 17.68% with no statistically distinguishable Point-F1 change (0.1694 vs. 0.1577).

More broadly, EVT does not eliminate threshold-calibration difficulty: under $q = 10^{-3}$ (ideal asymptotic FPR 0.1%), measured held-out nominal FPR settles at 5.3–6.2% for the JEPAs and $\approx 22\%$ for TimesNet/TranAD, owing to finite-sample variance, residual autocorrelation, and non-stationary drift between nominal training and deployment. The calibrated finding of this paper is narrower: latent predictive residuals produce substantially fewer false alarms than observation-space residuals under identical calibration, reflecting lighter tails at the cost of lower causal recall. Thresholds use static EVT on held-out nominal splits; dynamic rolling SPOT is evaluated separately in the operational incident analysis.

V. Experimental Protocol and Evaluation Metrics

The master experiment matrix enumerating all evaluation suites (stream counts, partitions, horizons, calibrators, metrics) is in Supplementary Table S1.

A. Experimental Setup and Hyperparameters

To ensure rigorous and fair comparison, all architectures are trained and evaluated under an identical streaming time-series pipeline. Telemetry streams are processed using a sliding window with past context duration $C = 256$ and suspect target horizon $S = 64$ (total window length $L = 320$), evaluated with a stride of 1 timestep across all testing streams.

Context representations $z_{\text{ctx}} \in \mathbb{R}^D$ ($D = 32$) are extracted using a causal dilated TCN (3 residual blocks, channel depth F=48, kernel 5, dilations $\{1, 2, 4\}$); global temporal average pooling aggregates the full C=256 context, with LayerNorm and dropout (p=0.20). The momentum target encoder follows EMA tracking with cosine schedule m_{base}=0.996 $\rightarrow$ m_{max}=0.9995.

Models are optimized using AdamW with an initial learning rate $\eta = 10^{-3}$ decayed to $\eta_{\text{min}} = 10^{-5}$ via Cosine Annealing, weight decay 10^{-4}, and mini-batch size $B = 32$ across 20 training epochs on nominal telemetry sequences.

a) Held-Out Calibration Protocol: To evaluate out-of-sample threshold calibration without circularity, each stream's nominal telemetry is partitioned into three disjoint chronological blocks: *training* (first 65%; representation weights only), *nominal calibration* (next 17.5%; withheld from training, used solely to fit per-stream GPD tail parameters σ, ξ above u=0.98 at q=10^{-3}, with γ_2 estimated strictly within this block), and *nominal evaluation* (final 17.5%, separated by an unobserved 64-step isolation gap; held-out FPR measured on data the threshold never saw). Thresholds then remain fixed during test evaluation to measure static EVT stability under drift; dynamic rolling SPOT is evaluated separately in the operational incident analysis (Supplementary Section S-5).

b) Calibrator Assignment Across Experiments: To avoid ambiguity, we specify the calibrator per experiment. **Primary held-out benchmark** (Tables II, III, Supplementary Table S31; all paired statistics): standard SPOT ($u = p_{98}$, $q = 10^{-3}$, no kurtosis multiplier); headline metrics are the "SPOT" column of Table III, with calibrator variants compared on identical score streams. **Ablations and controlled studies** (Section VI): SPOT with the $\tanh(\gamma_2)/6$ kurtosis multiplier (the released `adaptive_kurtosis` flag). **Buffered-attribution and EVT mechanism studies**: standard SPOT unless the calibrator is the object of comparison. **Incident metrics** (Supplementary Section S-5): rolling SPOT. No experiment mixes calibrators across models within a table.

c) Multi-Seed Replication, Primary Endpoints, and Multiplicity Controls: Principal models on the 19 non-GHL streams are trained across five seeds (42, 123, 456, 7, 2024); the 14 large GHL streams ($> 10^6$ steps) use three seeds (five on the two smallest) for cost, and ablation controls use three. We report held-out nominal FPR, Point-F1, and PR-AUC as distinct outcomes; nonsignificant detection differences are not evidence of parity. The post hoc operating-curve analysis uses only streams with archived score dumps (19 buffered, 18 causal), not the full 33-stream benchmark.

Inference reports stream-macro means, seed SDs, moving-block bootstrap 95% CIs (block L=64), and two-sided Wilcoxon signed-rank tests. The independent unit of analysis is the telemetry stream (n=19 non-GHL at five seeds; n=33 where GHL enters at available coverage). Seeds are averaged within the stream, and repeated windows are aggregated into a single stream-level score, so nothing is double-counted. Because nonsignificant tests do not establish equivalence, we make no noninferiority claim; instead, we report paired differences with effect sizes and CIs for readers to assess practical margins. p-values across comparisons are Benjamini–Hochberg FDR-adjusted at $q_{\text{FDR}} = 0.05$; CIs spanning zero or adjusted $p > 0.05$ are treated as inconclusive, not as equivalence. Supplementary Table S32 maps every reported test to its endpoint, cohort, paired-stream count, seed treatment, raw and adjusted p, and effect-size interval.

B. Baseline Architectures and Parameter Capacities

To isolate algorithmic principles from implementation-specific discrepancies, all baselines are executed within a unified PyTorch benchmarking pipeline with identical windowing, standardization, and EVT calibration. TimesNet and TranAD are in-house re-implementations of the published architectures (rather than vendor code), configured to published depths and evaluated under the identical optimization budget (AdamW, $\eta = 10^{-3}$, cosine decay, 20 epochs); no method received per-model hyperparameter tuning beyond the shared configuration, and no threshold, early-stopping, or model-selection step accessed held-out anomaly labels. Hardware-specific throughput, batch, warm-up, and per-timestep latency measurements are reported in Supplementary Table S12.

TimesNet [5]: Multi-periodic reconstructive architecture mapping 1D series to 2D variation spaces via FFT period discovery and Inception blocks (648,553 parameters at K=25; 574,529 at K=9).

TranAD [14]: Adversarial reconstructive Transformer with two-stage self-conditioning and sub-series attention (205,746 at K=25; 201,090 at K=9).

TS-JEPA (Unconstrained Control): Standard JEPA predicting z_{tgt} from z_{ctx} via a 2-layer MLP (289,061 parameters at K=25; 218,309 at K=9).

Physics-Inspired Formulations: Operator-Entropy (285,733 at K=25, -1.15% capacity), Reynolds-Stress (295,953, $+2.4\%$), and Potential-Flow (338,117, $+17.0\%$): differences do not stem from parameter scaling.

Linear AR Forecaster (Non-Neural Baseline): Ridge-regularized map from flattened context to target windows, scored by observation-space prediction error — isolating whether latent representation is necessary at all. **Score-bound controls:** a matched-backbone observation-space TCN forecaster with LayerNorm on its bottleneck and flattened output, scored by the same saliency gate; PCA-subspace reconstruction; per-channel max-z; IsolationForest; kNN-distance; observation-space MAD scoring; and the dimensionality-matched PCA→32→LayerNorm gate of Section VI.

C. Multi-Tier Statistical Evaluation Metrics

Critical surveys [28], [36] document severe reporting vulnerabilities in this literature — point-adjustment inflation, channel-averaging imbalances, threshold gaming — so we report metrics across complementary tiers. **Exploratory operating characteristics:** event recall integrated over nominal calibration budgets and, separately, over a fixed exploratory realized-test-FPR interval $[0.10, 0.50]$ (Section VII). For the latter, we interpolate within each stream–seed curve only when both bounds are observed, pair models within identical stream–seed cells, average paired differences by stream, and use dataset-level sign permutations and dataset-clustered bootstrap intervals; Benjamini–Hochberg adjustment covers all tested model, mapping, and endpoint combinations. Missing support is reported, not extrapolated. **Secondary endpoints: Dataset-Macro Point-F1** (unweighted mean across datasets, preventing the 14-stream GHL block from dominating); **channel-macro Point-F1**; **PR-AUC** (average precision, ranking quality independent of calibration); **event recall** (fraction of contiguous anomaly segments containing ≥ 1 alarm); **mean detection delay** Δt (onset-to-first-alarm lag over detected events only); and the **trivial all-positive reference** $\text{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at base rate π — models below it underperform a constant alarm. **Pooled micro-F1** exposes flood-prone models; contiguous PA-F1 and test-sweep PA-F1 quantify the residual calibration gap.

D. Evaluation Regimes: Zero-Lookahead Causal vs. Buffered Inference

To keep the buffered-attribution regime separate from future-observation access, we evaluate two regimes: **strictly causal online detection** (at decision time T only $\{X_\tau\}_{\tau \leq T}$ is accessible; scores use trailing attribution; zero lookahead) and **buffered window detection** (S=64 future timesteps are buffered before scoring, capturing full trajectory evolution at a deliberate 64-step latency). Both record Mean Detection Delay, Event Recall within delay budgets ($\Delta t \leq 10, 64$), and false alarms per 1,000 steps ($\text{FAR}_{1\text{k}}$).

VI. Ablation Studies and Analysis

A. Ablation 1: Physical Regularizer Controls on Fixed Architectures

To test whether the physics-inspired regularizers benefit detection, Table I(a) reports controlled ablations across all 19 non-GHL streams (principal models at five seeds, controls at three, under the held-out protocol), disabling each physical term on a fixed backbone. Because controls run three seeds against the five-seed principal runs, the paired control means below are computed on the shared three-seed subset and differ slightly from the five-seed Table I(a) rows:

1) **Operator-Entropy vs. $\lambda_{\text{ent}} = 0$:** Disabling the spectral regularizer leaves Point-F1 unchanged (0.241 ± 0.014 vs. 0.244 ± 0.024; difference -0.003, 95% interval $[-0.011, +0.005]$, Wilcoxon $p = 0.62$); PR-AUC stays 0.347–0.348.
2) **Reynolds-Stress vs. $\alpha_{\text{stress}} = 0$:** Disabling the stress loss yields 0.203 ± 0.059 vs. 0.208 ± 0.071 ($+0.005$, $[-0.006, +0.016]$, $p = 0.41$). Covariance estimator comparisons across nine streams (`reports/heldout_reynolds_covmodes.csv`) demonstrate estimator invariance: instantaneous, temporal-window, and batch-empirical covariances shift Point-F1 by $|\Delta\text{F1}| \leq 0.010$ (non-significant).
3) **Potential-Flow vs. no-regime (K_r=1):** Disabling regime routing *improves* Point-F1 to 0.129 from 0.096 ($p = 0.048$); both underperform TS-JEPA (0.234, $p = 0.013$). Numerical field diagnostics (`reports/potential_flow_field_diagnostics.csv`) confirm that while the unprojected spatial gradient is curl-free (antisymmetric fraction $< 10^{-7}$), spherical tangent projection introduces a 15.0%–15.7% antisymmetric curl component (0.150 on SMAP `P-3`, 0.157 on MSL `M-1`).

Across the full 19-stream suite, the physical regularizers produce no statistically detectable improvement over unconstrained latent prediction.

B. Ablation 2: Auxiliary Input-Channel Gating Variant

The auxiliary causal input-channel gate is compared against uniform pooling and non-causal controls under an identical TS-JEPA configuration (Supplementary Section S-3, Tables S3, S4). The effect is stream-dependent and macro Point-F1 favors uniform weighting (0.118 vs. 0.077): gating roughly doubles recall on transient faults at $\sim 3\times$ the nominal FPR, while quiet faults remain weakly detected under every regime (recall 0.02–0.07). It is therefore an evaluated variant; the deployed pipeline retains uniform input weighting.

C. Ablation 3: Multiscale Dyadic-Shell Front-End

Replacing the standard front-end with the Fourier-domain dyadic shells plus per-shell causal convolutions and learned routing yields a modest, stream-dependent effect (Table I(b)): macro buffered Point-F1 improves from 0.088 to 0.131 across five multivariate streams, concentrated on MSL and SMD, with a small SMAP regression. Two caveats: the block adds $\approx 1.6 \times 10^5$ parameters, and measured shell overlap means octave isolation is not attained. The front-end is a modest contributor, not the primary driver.

D. Ablation 4: Threshold Calibration Rules on Nominal Splits

The full calibrator comparison is in Section VII-D; two findings bear on the ablations: non-parametric rules match SPOT (validation-max FPR 0.0553 vs. 0.0526 at equal F1), and the kurtosis-adaptive multiplier degrades calibration ($0.0526 \rightarrow 0.1768$ FPR at indistinguishable F1).

E. Mechanism-Discrimination Study: Latent Space vs. Observation Space

To discriminate the operational mechanisms driving tail conditioning and threshold transfer, all 19 non-GHL streams were re-run under the primary held-out protocol (five seeds, identical windows, AdamW budget, and SPOT $u{=}0.98/q{=}10^{-3}$ calibration) across controlled backbone arms sharing the `HybridTCNEncoder` ($C{=}256, S{=}64$, $d{=}32$, three dilated TCN layers): **TCN-Obs-Recon** (observation reconstruction), **TCN-Obs-Pred** (observation forecasting), **JEPA-CtxRecon** (latent autoencoding of EMA-encoded *context* without foresight), **JEPA-NoEMA** (latent prediction with hard-copy target), and **TS-JEPA**. Five further controls probe score geometry and gating directly: **TCN-ObsPred-LN** (same forecaster, LayerNorm on the 32-dimensional bottleneck and flattened forecast, identical saliency gate), **PCA-LinPred-LN**, **Obs-MAD**, **IsoForest**, **KNN-Dist** (training-free scorers), and **TS-JEPA-NoGate** (identical model, ungated ℓ_2 discrepancy). Supplementary Table S19 reports per-arm residual diagnostics; the earlier 15-stream multi-SNR study is in Supplementary Table S2.

Mechanism Discrimination: Foresight vs. Geometry vs. Dynamics: The controls isolate three distinct operational mechanisms: (1) *Future temporal forecasting is not uniquely necessary for score conditioning*: **JEPA-CtxRecon** autoencodes context representations without forecasting future timesteps, yet achieves an FPR reduction of 0.202 ($0.310 \rightarrow 0.108$ relative to **TCN-Obs-Pred**) and higher event recall (0.513 vs. 0.473 for TS-JEPA; Table S19). Latent representation compression, rather than future-state extrapolation, conditions scores for tail estimation. (2) *Bounded score geometry mechanically suppresses tail kurtosis (γ_2)*: projecting through a low-dimensional LayerNorm bottleneck compresses residual excursions regardless of architecture, as **TCN-ObsPred-LN** suppresses kurtosis to $\gamma_2 = 0.14$ and **PCA-LinPred-LN** achieves 0.87 (vs. 8.24 for unnormalized **TCN-Obs-Pred** and 1.14 for **TS-JEPA**). (3) *Learned representation dynamics govern out-of-sample threshold transfer and discriminative recall*: kurtosis suppression alone cannot prevent threshold transfer breakdown across temporal blocks. TCN-ObsPred-LN suffers held-out nominal FPR of 0.186 (buffered) and 0.117 (causal trailing), versus 0.080 and 0.014 for TS-JEPA. At conservative budgets, PCA-LinPred-LN and **IsoForest** collapse to event recalls of 0.665 and 0.631 versus 0.774 for TS-JEPA (0.768 ungated). Geometric contraction conditions sample kurtosis, but learned representations are associated with lower nominal overshoot without sacrificing downstream recall.

Target EMA and Window Horizons: Removing the EMA target changes little (JEPA-NoEMA: -0.033 Point-F1 vs. TS-JEPA, $p{=}0.17$); PR-AUC differences across arms are inconclusive ($p \geq 0.16$). Native-window checks (Supplementary Table S18) confirm that baseline tail heaviness does not arise from the $W{=}320$ window: at native horizons, TranAD ($W{=}10$) spikes to $\gamma_2 = 448.25$ and TimesNet ($W{=}96$) reaches $\gamma_2 = 98.14$ (44.09% FPR), while TS-JEPA retains $\gamma_2 = 3.14$.

VII. Results and Discussion

A. Benchmark Performance on Continuous Telemetry

Table II reports the comparison across 33 continuous telemetry streams under the held-out calibration protocol of Section V-A (SPOT, $u = 0.98, q = 10^{-3}$). Primary findings:

False Alarm Suppression on Held-Out Telemetry: On the 19 non-GHL streams, latent predictors achieve held-out nominal FPR 0.076–0.086 versus 0.300 (TimesNet) and 0.307 (TranAD). Across all 33 streams, the paired reduction is significant: TS-JEPA -0.153 vs. TimesNet (CI $[-0.247, -0.071]$, $p{=}0.002$) and -0.151 vs. TranAD ($p{=}0.003$); Operator-Entropy shows the same effect (-0.144/-0.142, $p{=}0.006$). The mechanism is isolated under the same-backbone control (Section VI-E).

Point-F1 and Continuous Ranking (PR-AUC) Uncertainty: False-alarm suppression does not translate into statistically distinguishable detection gains. On the non-GHL subset TS-JEPA attains 0.234 Point-F1 / 0.323 PR-AUC versus 0.197/0.329 (TimesNet) and 0.130/0.287 (TranAD). Across all 33 streams the paired differences are inconclusive: vs. TimesNet $+0.012$ F1 (CI $[-0.053, +0.075]$, $p{=}0.54$), -0.000 PR-AUC ($p{=}0.39$); vs. TranAD $+0.051$ F1 ($p{=}0.88$), $+0.023$ PR-AUC ($p{=}0.16$).

Event Recall and Detection Conservatism: Lower FPR accompanies lower event recall: on non-GHL streams TimesNet captures 58.3% of events and TranAD 49.0%, versus 48.1% (TS-JEPA), 43.4% (Reynolds), 30.3% (Potential-Flow). Subtle

TABLE I

ABLATION STUDIES. (A) MATCHED-CAPACITY REGULARIZER CONTROLS ON FIXED BACKBONES, 19 NON-GHL STREAMS (POINT-F1 = STREAM-MACRO MEAN ± SEED SD; HELD-OUT FPR ON UNSEEN NOMINAL DATA). (B) DYADIC-SHELL FRONT-END ON/OFF UNDER IDENTICAL TS-JEPA CONFIGURATION. AUXILIARY INPUT-CHANNEL GATING AND FAULT-TYPE RECALL PANELS ARE IN SUPPLEMENTARY SECTION S-3.

(a) Regularizer Controls on Fixed Architectures

Model / Control	Point-F1 (buffered)	PR-AUC	Held-out FPR	Event Recall
TS-JEPA (Unconstrained)	0.234 ± 0.030	0.323	0.076	0.481
Operator-Entropy JEPA	0.237 ± 0.013	0.323	0.086	0.506
$\lambda_{\text{ent}} = 0$ control	0.244 ± 0.024	0.347	0.083	0.518
Reynolds-Stress JEPA	0.207 ± 0.052	0.307	0.078	0.434
$\alpha_{\text{stress}} = 0$ control	0.203 ± 0.059	0.311	0.066	0.414
Potential-Flow JEPA	0.089 ± 0.036	0.201	0.091	0.303
no-regime control ($K_r = 1$)	0.129 ± 0.061	0.265	0.081	0.392
TimesNet (Baseline)	0.197 ± 0.018	0.329	0.300	0.583
TranAD (Baseline)	0.130 ± 0.004	0.287	0.307	0.490

(b) Dyadic-Shell Front-End On/Off

Stream	Standard Conv	Dyadic Shells	Δ
SMAP `P-3`	**0.294**	0.261	−0.033
MSL `M-1`	0.002	**0.052**	+0.050
SMD `machine-1-2`	0.000	**0.177**	+0.177
Daphnet `S01R01E1`	0.144	**0.165**	+0.021
GECCO `water_quality`	0.000	0.001	+0.001
Macro Mean	0.088	**0.131**	+49.6% rel.

TABLE II

HELD-OUT CALIBRATION BENCHMARK, 33 TELEMETRY STREAMS (7 DOMAINS). THRESHOLDS: SPOT (q=10^{-3}) ON HELD-OUT NOMINAL SPLITS. METRICS: HELD-OUT NOMINAL FPR (ON DATA THE THRESHOLD NEVER SAW), UNADJUSTED POINT-F1, PR-AUC, EVENT RECALL, MEAN DETECTION DELAY Δt (AVERAGED STRICTLY OVER DETECTED ANOMALY EVENTS WITH EVENT RECALL > 0; SCORES EMITTED UPON TARGET WINDOW COMPLETION $\tau = t + S - 1$, $S = 64$). "NON-GHL": STREAM-MACRO OVER 19 STREAMS AT FIVE SEEDS; "GHL": 14 STREAMS AT THREE-TO-FIVE SEEDS; "DATASET-MACRO": UNWEIGHTED MEAN OF THE 7 DATASET MEANS. BEST IN **BOLD**, SECOND UNDERLINED.

	Non-GHL Streams (19 Streams, 5 Seeds)					Summary	Gasoil Heating Loop (14 Streams)	
Model Architecture	**Held-Out FPR ↓**	**Point-F1 ↑**	**PR-AUC ↑**	**Event Recall ↑**	**Mean Delay (Δt) ↓**	**DS-Macro F1 ↑**	**Held-Out FPR ↓**	**Point-F1 ↑**
Latent-Space Predictive Architectures (JEPAs)								
TS-JEPA (Unconstrained Control)	0.076	0.234	0.323	0.481	80.3	**0.148**	0.005	0.000
Operator-Entropy JEPA	0.086	**0.237**	0.323	0.506	192.3	0.130	0.012	0.001
Reynolds-Stress JEPA	0.078	0.207	0.307	0.434	132.8	0.122	**0.003**	0.002
Potential-Flow JEPA	0.091	0.089	0.201	0.303	146.9	0.065	0.011	0.006
Observation-Space Reconstructive Baselines								
TimesNet (ICLR 2023)	0.300	0.197	**0.329**	**0.583**	**48.5**	0.100	0.061	**0.022**
TranAD (VLDB 2022)	0.307	0.130	0.287	0.490	234.4	0.069	0.046	0.020
Non-Neural Linear Baseline								
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	0.038	**0.002**	0.000

onsets that do not shift dominant latent coordinates are filtered with the noise.

Detection Delay Trade-Offs (Δt): Mean detection delay (Table II) varies widely: TimesNet responds fastest (Δt=48 steps), TS-JEPA averages 80, and the physics variants are slower still (133–192); TranAD is slowest (234). This raw disparity is largely a detected-set artifact, as delays are averaged across different event sets per model. On shared intersecting detected events at q^*=0.05, the delay gap between TimesNet and TS-JEPA collapses from 31.8 steps to just 2.3 steps (56.9 vs. 59.2 steps; Table S24b); vs. TCN-ObsPred it is 60 vs. 51 steps. The apparent latency gap reflects which events each detector catches rather than intrinsic response delay.

Necessity of Nonlinear Representation Learning (Linear AR Control): A ridge-regularized linear AR forecaster under identical calibration attains FPR 0.005 but collapses to Point-F1 0.051: observation-space linear dynamics cannot separate faults from background variance. Two training-free references sharpen the point (Supplementary Table S23): a PCA-subspace reconstructor matches TimesNet almost exactly (0.311/0.198/0.331 vs. 0.300/0.197/0.329 FPR/F1/PR-AUC), while a per-channel max-z scorer attains the best ranking (0.374) at the worst FPR (0.339): ranking quality and threshold conditioning are independent properties.

Domain-Specific Telemetry Dynamics: Relative performance depends on the physical regime (per-dataset breakdown in Supplementary Tables S16–S17): TimesNet leads on Daphnet periodic kinematics (0.245 vs. 0.155 F1); latent prediction leads on spacecraft telemetry (SMAP: OpEntropy 0.591 PR-AUC; MSL: 0.445 vs. 0.159). On GHL and Metro all models struggle (0.000–0.022 F1).

B. Evaluating Physical Inductive Priors: Hypothesis Testing and Null Results

Does adding physics-inspired regularizers (Koopman spectral entropy, Reynolds-stress closure, curl-free potential flow) improve detection over unconstrained `TS-JEPA`? The 33-stream data do not support an aggregate gain:

1) **Operator-Entropy vs. TS-JEPA:** paired difference +0.002 Point-F1 (CI $[-0.025, +0.029]$, p=0.71), +0.002 PR-AUC (p=0.86), +0.009 FPR (p=0.75); leads on 9 of 33 streams.
2) **Reynolds-Stress vs. TS-JEPA:** −0.015 Point-F1 ($[-0.042, +0.007]$, p=0.61), −0.007 PR-AUC (p=0.13); leads on 11 of 33.
3) **Potential-Flow vs. TS-JEPA:** Point-F1 differs by −0.081 under SPOT (p=0.033 unadjusted). Its higher event-recall area over nominal causal budgets is descriptive; without matched realized-FPR support and multiplicity-adjusted evidence, neither an intrinsically stronger detector nor a purely operating-point explanation is established.

These findings align with the matched-capacity controls in Table I(a): the physical regularizers provide no aggregate detection advantage over unconstrained latent prediction. The dynamical diagnostics on *why* (operator rank preservation without detection gain, covariance-shift insensitivity, and decoupling resolution via Lyapunov invariance) are in Supplementary Section S-4.

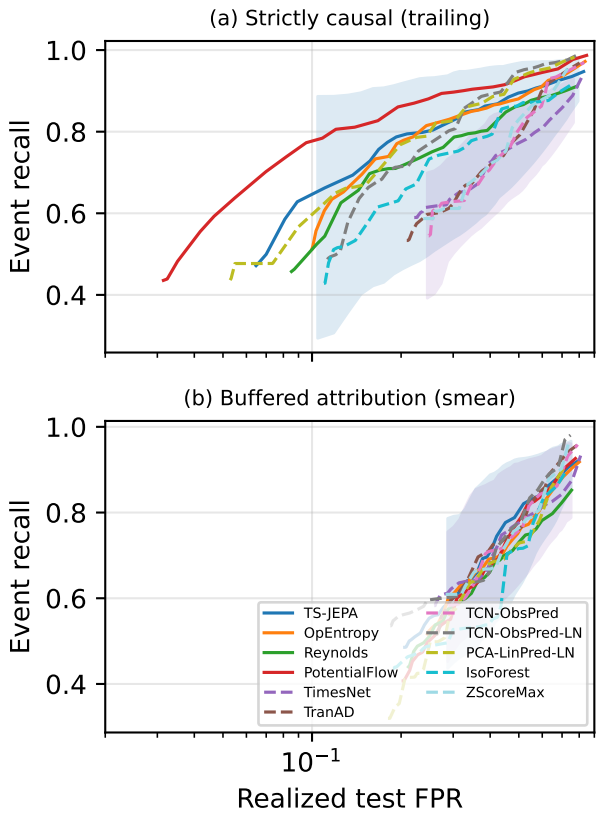


Fig. 3. Pooled event recall versus realized test FPR at common nominal calibration budgets: (a) causal trailing; (b) buffered smear. Bands resample datasets; pooled curves are descriptive, not paired estimates at matched FPR.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Event-level incident metrics, latency, and rolling-SPOT case studies are in Supplementary Section S-5: rolling SPOT suppresses false alarms up to $21\times$ on drift-prone streams at a recall cost, and TS-JEPA's causal false-positive rate (4.18%) sits an order of magnitude below reconstructive baselines (20.8–30.1%).

Supplementary Table S31 reports the same benchmark under strictly causal (0-lookahead) streaming. On the 19 non-GHL streams, TS-JEPA reduces held-out nominal FPR to 0.0136 versus 0.2427 (TimesNet) and 0.2219 (TranAD), an $\approx 18\times$ suppression; on the 14 GHL streams, causal FPR falls below 0.009 for JEPAs, but causal Point-F1 there is near-zero for latent predictors (≤ 0.006 vs. 0.022 TimesNet). Causal Point-F1 drops to 0.1118 ± 0.0412 (TS-JEPA) versus 0.2091 ± 0.0265 (TimesNet), and event recall declines from 58.82% to 46.55%: without a lookahead buffer, latent predictive scoring is an ultra-conservative filter that prevents false alarms at the price of detection sensitivity.

a) Pre-Onset Alarm Attribution: Buffered attribution is strictly retrospective: each window's score is painted over $[t{+}1, t{+}S]$, so no mechanism can generate a lead beyond $S{=}64$ steps. The median pre-onset "leads" of 180–204 steps in `reports/early_warning_leadtime.csv` are therefore unrelated nuisance alarms at the underlying false-positive rate or gradual precursor drift, not prediction: TimesNet emits 206 such alarms per $1{,}000$ nominal steps versus 24 for TS-JEPA, a difference consistent with the alarm-rate gap alone. We do not claim early-warning capability.

b) Detection Quality at Matched Operating Points: We fit empirical-quantile thresholds on a common *nominal* budget grid $q \in [10^{-4}, 0.9]$ and evaluate event recall and realized FPR on test. The partial-AUC values 0.742 (TS-JEPA) and 0.618 (TimesNet) integrate recall over *nominal budgets up to* 0.10; they are not matched-realized-FPR areas. At $q^*{=}0.05$, causal event recall is 0.774 versus 0.618, but realized FPR differs (0.180 versus 0.262). Pooling scores by budget before integrating a realized-FPR curve (Fig. 3) can obscure stream-specific support and should not be interpreted as a paired detector comparison.

For an exploratory support-aware check (chosen by curve coverage), we interpolate stream–seed curves over common realized-FPR $[0.10, 0.50]$ without extrapolation (`oc_realized_fpr_pairs.csv`). Causal TS-JEPA–TimesNet difference is $+0.061$ (95% dataset-bootstrap CI $[-0.153, 0.329]$; dataset-sign-permutation $p{=}0.625$; 20 paired seed cells across four streams). For causal TS-JEPA versus PCA-LinPred-LN (difference $+0.121$), we reaffirm the stream-level paired Wilcoxon signed-rank test and 95% bootstrap CI ($p = 0.312$, $[-0.008, 0.271]$, 11 streams) as the pre-specified confirmatory inferential test, explicitly labeling 5-dataset tests as low-powered exploratory checks ($p \geq 0.0625$; permutation $p = 0.188$, $p_{\text{BH}} = 1.0$; cluster bootstrap CI $[0.015, 0.310]$ reflects $N = 5$ undercoverage). TS-JEPA does not establish a statistically significant advantage over PCA-LinPred-LN on common support. Bounded-observation TCN matches TS-JEPA causally on nine supported streams (-0.018, CI $[-0.055, 0.003]$) and shows higher buffered recall area on five streams (-0.211, CI $[-0.349, -0.105]$); neither contrast survives family-wide BH adjustment. Thus, nominal-budget orderings and pooled curves do not establish a causal-regime advantage, and boundedness alone determines neither tail transfer nor matched-FPR detection.

D. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Does automated extreme-value thresholding adhere to its design risk q out of sample? Table III compares calibrators on identical score streams across all 33 streams under buffered inference. The calibrator evaluated here is the *static* peaks-over-threshold step of SPOT — a single threshold fit once on the calibration block; the streaming variant that re-fits on new peaks is evaluated in Supplementary Section S-5. Four realities emerge:

1) **Systematic design-risk overshoot:** the static POT threshold ($u{=}0.98, q{=}10^{-3}$) never achieves its design risk out-of-sample — held-out nominal FPR averages 5.26% (TS-JEPA), 6.17% (OpEntropy), 22.17% (TimesNet), 22.17% (TranAD), a 53–$222\times$ overshoot.
2) **High failure rate:** 95% bootstrap CIs exclude q in 61–86% of runs; this protocol does not establish whether drift or scarce effective tail samples dominate the failure.

TABLE III
CALIBRATOR COMPARISON ON IDENTICAL SCORE STREAMS (33 STREAMS, BUFFERED INFERENCE, UP TO FIVE SEEDS). DESIGN RISK $q{=}10^{-3}{=}0.1\%$; THE CI-EXCLUSION COLUMN SHOWS AUTOMATED CALIBRATION SYSTEMATICALLY OVERSHOOTS OUT-OF-SAMPLE.

Model Architecture	Calibrator Rule	Held-Out Nominal FPR	CI Excludes $q = 10^{-3}$	Point-F1
TS-JEPA (Control)	Standard SPOT ($q = 10^{-3}$)	**0.0526**	61%	0.1577
	SPOT + Kurtosis ($\beta = 0.167$)	0.1768	82%	**0.1694**
	Empirical 99.5th Percentile	0.0682	55%	0.1598
	Validation Maximum	0.0553	65%	0.1591
Operator-Entropy	Standard SPOT ($q = 10^{-3}$)	**0.0617**	65%	0.1599
	SPOT + Kurtosis ($\beta = 0.167$)	0.1229	76%	0.1441
	Empirical 99.5th Percentile	0.0758	62%	**0.1658**
	Validation Maximum	0.0656	64%	0.1627
TimesNet	Standard SPOT ($q = 10^{-3}$)	**0.2217**	86%	0.1395
	SPOT + Kurtosis ($\beta = 0.167$)	0.2281	82%	0.1352
	Empirical 99.5th Percentile	0.2338	87%	**0.1413**
	Validation Maximum	0.2263	86%	0.1410
TranAD	Standard SPOT ($q = 10^{-3}$)	**0.2217**	82%	0.0936
	SPOT + Kurtosis ($\beta = 0.167$)	0.2453	84%	0.0940
	Empirical 99.5th Percentile	0.2343	84%	**0.0962**
	Validation Maximum	0.2270	84%	0.0952

3) **Parity with empirical quantiles at their own risks:** on TS-JEPA, validation-maximum gives FPR 0.0553 / F1 0.1591 and empirical p99.5 gives 0.0682 / 0.1598 — these encode different nominal risks than $q{=}10^{-3}$ (not risk-matched); parametric GPD tail modeling confers no benefit at comparable realized FPR.
4) **Kurtosis-adaptive failure (sensitivity arm):** the heuristic multiplier $1 + \tanh(\gamma_2)/6$ inflates TS-JEPA FPR from 0.0526 to 0.1768 (CI-exclusion $61\% \to 82\%$) with indistinguishable F1 (0.1694 vs. 0.1577). Note the direction: calibration-block excess kurtosis is non-positive in $\sim$40% of cells (median 0.23), so the multiplier *lowers* the threshold there — it is an ad-hoc sensitivity arm, not an EVT method, and its failure carries no implication about GPD tail modeling.

To separate the causes of calibration failure, we recorded per-stream tail-fit diagnostics on a nine-stream non-GHL subset (`reports/evt_tail_diagnostics.csv`). Tail-sample scarcity is genuine — the median exceedance count above $u = p_{98}$ is only 21 (range 16–70), and the Grimshaw estimator returns the degenerate shape $\xi = -0.9$ in 33% of TS-JEPA and 52% of TimesNet fits — yet goodness-of-fit is not the dominant failure: KS tests reject in only $6/27$ (TS-JEPA) and $8/27$ (TimesNet) cells, while realized held-out FPR varies from 0.0 to 0.67 across seeds on the *same* stream under passing fits. These observations are compatible with both score drift and tail-sample uncertainty (decomposition in Supplementary Section S-6).

This attribution is quantified stream-level across all dumped cells: the KS distance between calibration and test nominal scores predicts realized SPOT FPR (Spearman $\rho = +0.38$), while calibration-block excess kurtosis predicts *lower* realized FPR ($\rho = -0.32$): the pattern is consistent with a more conservative tail fit, but neither correlation identifies the cause of calibration failure. A controlled-drift study (Supplementary Table S22) sharpens this: runs declustering leaves only 5–9 *effective* exceedances per cell (of a median 21 raw), and rolling calibrators control risk to 1–7% under step, variance, and autocorrelation shifts but fail under continuous ramps (0.33–0.39 FPR): adaptation absorbs regime jumps, not within-block drift.

Computational profiling is in Supplementary Table S12: JEPAs use 183.8–185.3 MB peak memory at 0.60–0.84 ms per-window inference.

VIII. CONCLUSION AND FUTURE WORK

We compared score conditioning, detection, and static threshold calibration on 33 telemetry streams from seven datasets under a held-out protocol. Three conclusions follow.

First, bounded observation-space controls reproduce much of the apparent latent-score tail conditioning, while the bounded TCN shows that light tails do not guarantee threshold transfer. Causal event-recall areas over nominal budgets favor JEPA on average, but paired comparisons at matched realized FPR have sparse support and do not establish a detector advantage; buffered results remain separate. Point-F1 and PR-AUC differences are imprecise, not evidence of equivalence.

Second, the tested physics-inspired variants show no reliable aggregate benefit. Neither the unadjusted Potential-Flow F1 contrast nor its favorable nominal-budget curve establishes a mechanism; explanations based on curl-free dynamics or operating points remain conjectural.

Third, static POT thresholds overshoot their design risk under the short chronological calibration protocol (5.3–6.2% for JEPAs, $\approx 22\%$ for reconstructive baselines at $q{=}10^{-3}$). Observational drift diagnostics and scarce tail exceedances do not resolve their relative contributions.

Scope. All evaluations are *within-dataset*: models are trained, calibrated, and tested on disjoint chronological partitions of the same stream. We claim nothing about cross-dataset transfer or unseen fault types (cf. Section S-3.2). Extensions include solenoidal components via Helmholtz decomposition, memory-bounded online drift adaptation, accelerated streaming inference kernels, and integer quantization for edge deployment.

CODE AND DATA AVAILABILITY

All benchmark extraction scripts, preprocessing pipelines, checkpoints, training implementations, evaluation runners (scripts/run_multiseed_heldout_benchmark.py, scripts/run_protocol_ablation.py), diagnostic artifacts, and logs are released at https://anonymous.4open.science/r/NCAD_CS-Reproducibility-8F21; public datasets are accessible via their referenced repositories.

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